

Lab04 – Ephemeral Volume

In this exercise, you will create a Pod that runs the web server nginx. Nginx requires certain directory paths to be writable. We'll mount ephemeral Volumes to make those paths available to container.

1. Create a Pod named `nginx` in the namespace `h92`. Its container should run the container image `nginx:1.21.6`.

```
brahim@Training:~/Lab04$ kubectl create ns h92
namespace/h92 created
brahim@Training:~/Lab04$ vim nginx-pod.yaml
brahim@Training:~/Lab04$ cat nginx-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: h92
spec:
  containers:
  - name: nginx
    image: nginx:1.21.6

brahim@Training:~/Lab04$
brahim@Training:~/Lab04$ kubectl apply -f nginx-pod.yaml
pod/nginx created
brahim@Training:~/Lab04$
brahim@Training:~/Lab04$ □
```

2. Define the Volumes of type `emptyDir` named `nginx-run`, `nginx-data` and `nginx-cache` with their corresponding mount paths `/var/run`, `/usr/local/nginx` and `/var/cache/nginx`. The modified YAML manifest could look as shown below.

```
brahim@Training:~/Lab04$ vim nginx-pod.yaml
brahim@Training:~/Lab04$ cat nginx-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: h92
spec:
  containers:
  - name: nginx
    image: nginx:1.21.6
    securityContext:
      volumeMounts:
      - name: nginx-run
        mountPath: /var/run
      - name: nginx-cache
        mountPath: /var/cache/nginx
      - name: nginx-data
        mountPath: /usr/local/nginx
  volumes:
  - name: nginx-run
    emptyDir: {}
  - name: nginx-data
    emptyDir: {}
  - name: nginx-cache
    emptyDir: {}

brahim@Training:~/Lab04$ kubectl delete pod nginx -n h92 --force
Warning: Immediate deletion does not wait for confirmation that the running resource
the cluster indefinitely.
pod "nginx" force deleted
brahim@Training:~/Lab04$ kubectl apply -f nginx-pod.yaml
pod/nginx created
brahim@Training:~/Lab04$ □
```

3. Say you would want to ensure the nginx can only write to those Volume mount paths but not the container's temporary file system. How do you prevent this from being allowed?

```
brahim@Training:~/lab04$ vim nginx-pod.yaml
brahim@Training:~/lab04$ cat nginx-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: h92
spec:
  containers:
  - name: nginx
    image: nginx:1.21.6
    securityContext:
      readOnlyRootFilesystem: true
  volumeMounts:
  - name: nginx-run
    mountPath: /var/run
  - name: nginx-cache
    mountPath: /var/cache/nginx
  - name: nginx-data
    mountPath: /usr/local/nginx
  volumes:
  - name: nginx-run
    emptyDir: {}
  - name: nginx-data
    emptyDir: {}
  - name: nginx-cache
    emptyDir: {}
brahim@Training:~/lab04$ kubectl delete pod nginx -n h92 --force
Warning: Immediate deletion does not wait for confirmation that the running resource has been deleted
pod "nginx" force deleted
brahim@Training:~/lab04$ kubectl apply -f nginx-pod.yaml
pod/nginx created
brahim@Training:~/lab04$
```